FDIT 2230 · Spring 2026 · P02-A

YouTube Trending Videos

From Big Data to Interaction — P02-A First Deliverable

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Dataset
YouTube Trending

DATASET OVERVIEW

What Is This Dataset?

40,000+	16	10	2017–18
Rows (approx.)	Columns	Countries	Year Range

Key Columns

video_id
Unique identifier per video

trending_date
Date when the video trended

title
Video title (text)

channel_title
Uploader channel name

category_id
Content category (mapped to label)

views / likes / dislikes / comment_count
Engagement metrics (numeric)

tags
Pipe-separated tags string

thumbnail_link
URL to video thumbnail

country
One of: US, GB, CA, DE, FR, IN, JP, KR, MX, RU

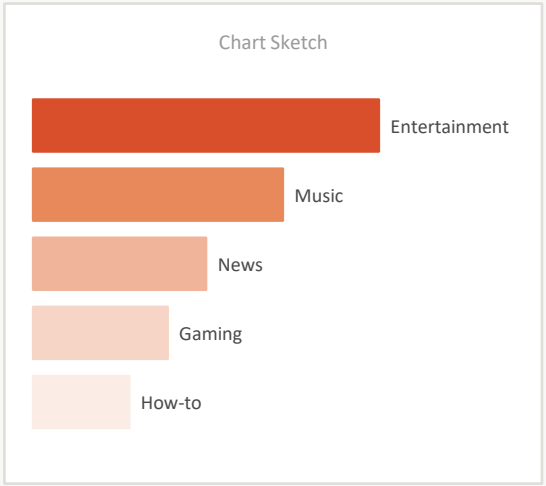
STORY 01 OF 03

What Content Categories Dominate Trending?

Not all video categories trend equally. This story explores which content types Entertainment, Music, News, Gaming, How-to consistently make it to the trending page, and whether that distribution varies by country. The goal is to reveal structural bias in what gets amplified on the platform.

Planned Interaction

Stacked bar chart showing category share per country. Hovering a segment reveals exact count and percentage. A **country filter** allows switching between regions for direct comparison.



Navigation: Overview dashboard → drill-down by category → individual channel detail

STORY 02 OF 03

How Long Does It Take a Video to Trend?

The dataset contains both the publish date and the trending date. Calculating the gap between them reveals how quickly different categories and channels break into the trending list and whether virality follows a pattern or is unpredictable.

Planned Interaction

Still confused for the chart for this panel.

Chart Sketch

Navigation: Overview dashboard → drill-down by days-to-trend range → video detail view

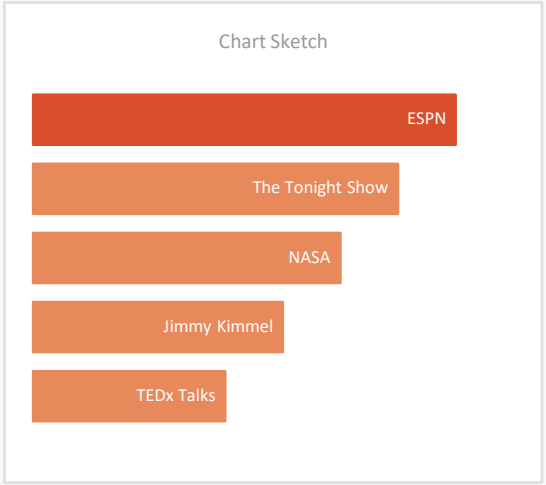
STORY 03 OF 03

Which Channels Trend Repeatedly - and Why?

Some channels appear on the trending list dozens of times. This story identifies repeat trendsetters, visualizes their engagement rates (likes-to-views ratio), and examines what separates them from one-time trending videos. It reveals whether trending is earned or algorithmic.

Planned Interaction

Horizontal ranked bar chart of top 20 channels by trending appearances. Clicking a channel expands a **detail panel** showing engagement breakdown over time (line chart), enabling overview-to-detail navigation.



Navigation: Overview dashboard → ranked channel list → per-channel engagement detail

INTERACTION DESIGN

Navigation Between Views

Overview Dashboard

5+ KPI indicators
Category distribution
Country selector

Drill-Down A

Category deep-dive
Stacked bar + filter
Hover info cards

Drill-Down B

Channel ranking
Engagement timeline
Click to expand

Hover info cards reveal title, views, and likes · Country / category filter updates all charts · Click on bar/bubble navigates to detail view · Back button returns to overview

DATA EXPLORATION

Initial Findings & Cleaning Plan

01 Duplicate Rows

Same video_id appears multiple times (once per trending day). Will aggregate by video_id — summing trending appearances, keeping max views.

02 category_id is Numeric

Raw data uses integer IDs. Will join with the category JSON file per country to decode labels (e.g. 24 → Entertainment).

03 Inconsistent Tags

Tags are pipe-separated strings or '[none]'. Will parse into arrays; '[none]' treated as null for tag-based analysis.

04 Missing publish_time

Some rows lack publish_time, preventing days-to-trend calculation. Will drop these rows only for Story 02; other stories retain full dataset.

Source: Kaggle – YouTube Trending Video Dataset (Mitchell J, 2019) · ~40,000 rows × 16 columns across 10 countries

NEXT STEPS

Toward P02-B

Week 09

Data Structuring & Cleaning

Merge country files, decode category_id, compute days-to-trend, remove duplicates, export clean CSV (~40k rows)

Week 10

P02-A Submission

1:1 presentation + raw dataset upload

Week 12

Studio — Dashboard Build

Implement 3-page interactive dashboard in D3.js or Chart.js: overview + 2 drill-downs, hover + filter interactions

Week 13

P02-B Due

Final in-class presentation with live dashboard demo, clean dataset, and PDF slide deck